

3.3.2 System Testing

A series of tests was conducted to ensure the web-based timesheet system operates correctly and meets the required specifications as shown in Table 4.

Table 4 System Testing Plan

Test Case ID	Test Scenario	Expected Result
TC-01	Log In	User successfully logged into the system.
TC-02	Record Timesheet	User records daily task and project details, including start and end times.
TC-03	Edit Timesheet	User can modify existing entries before final submission.
TC-04	Submit Timesheet	User can submit their finalized timesheet directly in the system.
TC-05	View Timesheet	Users are able to see their edited entries before and after submission.
TC-06	Track Submission Status	Enables users to track the submission status of timesheets.
TC-07	Manage User Accounts	Users can update their personal information and change their passwords.
TC-08	Manage Projects	User can add, edit or remove project.
TC-09	Set System Configurations	User can configure settings such as submission deadlines.
TC-10	Log Out	Users are able to securely log out of the system.

RESULTS AND DISCUSSION

This section discusses the outcomes of the system development and testing phases, analyzing how the results meet the project requirements and resolve the issues identified in the existing system.

4.1 Project Achievement

The Strato Web-Based Timesheet System successfully fulfills the primary functional requirements defined during the design phase. The project achieved its core objectives by replacing the company's manual, Excel-based reporting management process with a standardized digital platform. By centralizing the submission process, the system will eventually eliminate the need for manual email-based management. This allows staff to submit and track their timesheet in a single location, providing a standardized data source for future HR review once the administrative modules are fully integrated.

4.2 Testing and Evaluation Results

The results of testing and evaluating the system's performance and functionality are shown in Table 5.

Table 5 Testing Result

Test Case ID	Test Scenario	Expected Result	Status
TC-01	Log In	User successfully logged into the system.	Pass
TC-02	Record Timesheet	User records daily task and project details, including start and end times.	Pass
TC-03	Edit Timesheet	User can modify existing entries before final submission.	Pass
TC-04	Submit Timesheet	User can submit their finalized timesheet directly in the system.	Pass
TC-05	View Timesheet	Users are able to see their edited entries before and after submission.	Pass
TC-06	Track Submission Status	Enables users to track the submission status of timesheets.	Pass
TC-07	Manage User Accounts	Users can update their personal information and change their passwords.	Pass
TC-08	Manage Projects	User can add, edit or remove project.	Pass
TC-09	Set system configurations	User can configure settings such as submission deadlines.	Pass
TC-10	Log Out	Users are able to securely log out of the system.	Pass

Based on the testing results, all test cases were successfully executed and passed. This indicates that the system is functioning as intended and meets the specified requirements. The results demonstrate that the system is stable and reliable.

4.3 Stakeholder Feedback

Following the technical demonstration, feedback was gathered from the supervisor and the IT team. Their assessment focused on the system's usability and its potential to solve existing issues. The stakeholders highlighted the automated pre-filling of data such as dates, months, year and positions as a significant improvement. They noted that by automating these fields, the system effectively eliminates common human error and maintains data standardization across all entries. This allows users to focus solely on

documenting their projects and tasks, thereby increasing administrative efficiency and ensuring the submitted data remains consistent for future reporting.

Additionally, the feedback emphasized the importance of automated submission status tracking, such as 'Pending', 'Submitted' and 'Late Submission', which shows staff exactly which timesheets they still need to submit. This feature directly addresses the recurring issue where staff would previously forget to submit their timesheets. This feature facilitates a more disciplined reporting process, ensuring that all entries are finalized and submitted in accordance with the company deadlines.

Furthermore, security was a key focus during the feedback session. The system uses generic error messages, such as 'Credentials do not match our records', to prevent unauthorized users from identifying valid email addresses. The system also includes a strong password policy and login attempt limits to protect against brute-force attacks. This feedback confirmed that the system offers a more secure way to manage employee data than the previous manual method.